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Tricuspid valve repair devices and associated systems and methods

Abstract

Tricuspid valve repair devices and associated systems and methods are disclosed herein. A tricuspid valve repair device configured in accordance with embodiments of the present technology can include, for example, a coaptation member configured to be positioned between one or more native leaflets of the tricuspid valve to at least partially fill a space between the native leaflets. The tricuspid valve repair device can further include one or more fixation mechanisms for securing the coaptation member in position between the leaflets. The fixation mechanisms can include clip mechanisms, lock mechanisms, stabilization members, anchors, and/or other structures configured to engage cardiac anatomy local to or remote from the tricuspid valve, such as the native leaflets, the tricuspid valve annulus, the right ventricular outflow tract, the superior vena cava, the inferior vena cava, and so on.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/116,729, titled “TRICUSPID VALVE REPAIR DEVICES AND ASSOCIATED SYSTEMS AND METHODS,” and filed Nov. 20, 2020, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) The present technology is directed to devices, systems, and methods for repairing a tricuspid valve, and more particularly to valve repair devices configured to be implanted at the tricuspid valve.

BACKGROUND

(2) Proper functioning of the tricuspid valve of a human patient can be affected by valve regurgitation, valve prolapse, and/or valve stenosis. Tricuspid valve regurgitation can occur when the native leaflets of the tricuspid valve fail to coapt into apposition at peak contraction pressures such that blood leaks past the valve from the right ventricle past the tricuspid valve and into the right atrium. Several structural factors may affect the proper closure of the tricuspid valve leaflets. For example, an enlarged tricuspid valve annulus caused by dilation of heart muscle may prevent

proper coaptation of the leaflets during systole. Other conditions involve a stretch or tear in the chordae tendineae—the tendons connecting the papillary muscles to the tricuspid valve leaflets—which may also affect proper closure of the valve leaflets. A ruptured chordae tendineae, for example, may cause a valve leaflet to prolapse (e.g., abnormally bulge up) into the right atrium due to inadequate tension on the leaflet, which can also lead to valve regurgitation. Abnormal backflow can also occur when the papillary muscles are compromised (e.g., due to ischemia) such that the affected papillary muscles do not contract sufficiently to effect proper closure during systole. Normal tricuspid valve functioning can also be affected by valve stenosis (e.g., a narrowing of the valve orifice) which, for example, can impede filling of the right ventricle during diastole.

(3) Tricuspid valve regurgitation is often treated using diuretics and/or vasodilators to reduce the amount of blood flowing back into the right atrium. Other treatment methods, such as surgical approaches (open and intravascular), have also been used to either repair or replace native tricuspid valves. For example, cinching or resecting portions of the dilated annulus are typical repair approaches. Cinching of the annulus has been accomplished by implanting annular or peri-annular rings which are generally secured to the annulus or surrounding tissue. Other repair procedures have also involved suturing or clipping of the valve leaflets into partial apposition with one another. Alternatively, more invasive procedures replace the entire valve with mechanical valves or biological tissue. These invasive procedures are conventionally done through large open thoracotomies and are thus very painful, have significant morbidity, and require long recovery periods.

(4) However, with many repair and replacement procedures, the durability of the devices or improper sizing of annuloplasty rings or replacement valves may cause complications. Moreover, many of the repair procedures depend upon the skill of the cardiac surgeon since poorly or inaccurately placed sutures may affect the success of procedures.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Many aspects of the present disclosure can be better understood with reference to the following drawings. The components in the drawings are not necessarily to scale. Instead, emphasis is placed on clearly illustrating the principles of the present disclosure.

(2) FIG. 1 is a diagram of a tricuspid valve and surrounding anatomy at which a tricuspid valve repair device can be implanted in accordance with embodiments of the present technology.

(3) FIGS. 2A-2D are a front view, a side view, a rear view, and a top view, respectively, of a tricuspid valve repair device that can be implanted in a heart of a subject in accordance with embodiments of the present technology.

(4) FIGS. 3A-3D are a front view, a side view, a rear view, and a top view, respectively, of the valve repair device of FIGS. 2A-2D with a stabilization member omitted in accordance with embodiments of the present technology.

(5) FIGS. 4A-4D are a front view, a side view, a rear view, and a top view, respectively, of the valve repair device of FIGS. 2A-2D including a flexible, generally-round stabilization member in accordance with embodiments of the present technology.

(6) FIGS. 5A and 5B are a side view and a rear view, respectively, of the valve repair device of FIGS. 2A-2D including an elongate-curved stabilization member in accordance with embodiments of the present technology.

(7) FIGS. 6A-6H are transverse cross-sectional views of various coaptation members in accordance with embodiments of the present technology.

(8) FIGS. 7A-7L are side cross-sectional views of various coaptation members in accordance with embodiments of the present technology.

(9) FIG. 8A is a side cross-sectional view of a valve repair device implanted at a tricuspid valve in accordance with embodiments of the present technology. FIGS. 8B and 8C are transverse cross-sectional views of the valve repair device of FIG. 8A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(10) FIG. 9A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 9B and 9C are transverse cross-sectional views of the valve repair device of FIG. 9A during diastole and systole, respectively, in accordance with embodiments of the present technology. FIG. 9D includes several side cross-sectional views of a coaptation member of the valve repair device of FIGS. 9A-9C in accordance with embodiments of the present technology.

(11) FIG. 10A is a top view of a valve repair device in accordance with embodiments of the present technology. FIG. 10B is a side cross-sectional view of the valve repair device of FIG. 10A implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 10C and 10D are transverse cross-sectional views of the valve repair device of FIGS. 10A and 10B during diastole and systole, respectively, in accordance with embodiments of the present technology.

(12) FIG. 11A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 11B and 11C are transverse cross-sectional views of the valve repair device of FIG. 11A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(13) FIG. 12A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 12B and 12C are transverse cross-sectional views of the valve repair device of FIG. 12A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(14) FIG. 13A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 13B and 13C are transverse cross-sectional views of the valve repair device of FIG. 13A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(15) FIG. 14A is a front view of a tricuspid valve repair device in accordance with embodiments of the present technology. FIG. 14B is a side cross-sectional view of the valve repair device of FIG. 14A implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 14C and 14D are transverse cross-sectional views of the valve repair device of FIGS. 14A and 14B during diastole and systole, respectively, in accordance with embodiments of the present technology. FIG. 14E is a front view of the tricuspid valve repair device of FIGS. 14A-14D in accordance with additional embodiments of the present technology.

(16) FIG. 15A is a top view of a valve repair device in accordance with embodiments of the present technology. FIG. 15B is a side cross-sectional view of the valve repair device of FIG. 15A implanted at the tricuspid valve in accordance with embodiments of the present technology.

(17) FIG. 16A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 16B and 16C are transverse cross-sectional views of the valve repair device of FIG. 16A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(18) FIGS. 17A, 18A, and 19A are side cross-sectional views of valve repair devices implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 17B, 18B, and 19B are transverse cross-sectional views of the valve repair devices of FIGS. 17A, 18A, and 19A, respectively, during diastole in accordance with embodiments of the present technology. FIGS. 17C, 18C, and 19C are transverse cross-sectional views of the valve repair devices of FIGS. 17A, 18A, and 19A, respectively, during systole in accordance with embodiments of the present technology.

(19) FIG. 20A is a side cross-sectional view of a valve repair device implanted at the tricuspid

valve in accordance with embodiments of the present technology. FIGS. 20B and 20C are transverse cross-sectional views of the valve repair device of FIG. 20A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(20) FIG. 21A is a side cross-sectional view of the valve repair device of FIGS. 20A-20C including an atrial stabilization member and implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 21B and 21C are transverse cross-sectional views of the valve repair device of FIG. 21A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(21) FIG. 22 is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology.

(22) FIG. 23 is a side cross-sectional view of the valve repair device of FIG. 22 implanted at the tricuspid valve and secured to the anterior and/or posterior leaflets of the tricuspid valve in accordance with embodiments of the present technology.

(23) FIGS. 24A and 24B are side cross-sectional views of a valve repair device implanted at the tricuspid valve in a first position and a second position, respectively, in accordance with embodiments of the present technology.

(24) FIG. 25A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 25B and 25C are transverse cross-sectional views of the valve repair device of FIG. 25A during diastole and systole, respectively, in accordance with embodiments of the present technology.

(25) FIG. 26 is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology.

(26) FIG. 27A is a side cross-sectional view of a valve repair device implanted at the tricuspid valve in accordance with embodiments of the present technology. FIGS. 27B and 27C are transverse cross-sectional views of the valve repair device of FIG. 27A during diastole and systole, respectively, in accordance with embodiments of the present technology.

DETAILED DESCRIPTION

(27) The present technology is directed to tricuspid valve repair devices and associated systems and methods. In some embodiments, for example, a tricuspid valve repair device (also referred to herein as a “valve repair device,” “coaptation assist device,” “implant device,” and iterations thereof) includes features that anchor to native anatomy of a heart of a patient, such as one or more of the native leaflets of the tricuspid valve of a human patient. For example, the tricuspid valve repair device can include (i) a coaptation member (also referred to as a “coaptation structure,” “space filler,” “filler,” “baffle,” “intravalvular body,” “intermediate structure,” and iterations thereof) positioned at least partially between the native tricuspid valve leaflets, and (ii) one or more clip mechanisms that secure the coaptation member in position relative to the native leaflets. The coaptation member can at least partially fill a regurgitant orifice in the tricuspid valve and provide a new coaptation surface for the native leaflets to seal around. The coaptation member can also push a portion of the native leaflets outward toward the right ventricular wall, while reducing or minimizing disruption of the remaining portion of the native leaflets. The clip mechanisms can engage the ventricular and/or the atrial side of the native leaflets to secure the position of the coaptation member relative to the tricuspid valve.

(28) In some embodiments, a tricuspid valve repair device in accordance with additional embodiments of the present technology can include (i) a coaptation member positioned between the native tricuspid valve leaflets, and (ii) one or more anchors and/or brace members that secure the coaptation member to anatomy of the right heart other than the native tricuspid valve leaflets. For example, the anchors can be secured to the right atrial wall, right ventricular wall, tricuspid valve annulus, one or more tricuspid leaflets, right ventricular outflow tract, inferior vena cava, superior vena cava, coronary sinus, and/or other portions of the anatomy of the right heart of the patient.

(29) Specific details of several embodiments of the present technology are described herein with

reference to FIGS. 1-27C. The present technology, however, can be practiced without some of these specific details. In some instances, well-known structures and techniques often associated with catheter-based delivery systems, prosthetic tricuspid heart valves, and the like, have not been shown in detail so as not to obscure the present technology. The terminology used in the description presented below is intended to be interpreted in its broadest reasonable manner, even though it is being used in conjunction with a detailed description of certain specific embodiments of the disclosure. Certain terms can even be emphasized below; however, any terminology intended to be interpreted in any restricted manner will be overtly and specifically defined as such in this Detailed Description section.

(30) The accompanying Figures depict embodiments of the present technology and are not intended to be limiting of its scope. The sizes of various depicted elements are not necessarily drawn to scale, and these various elements can be arbitrarily enlarged to improve legibility. Component details can be abstracted in the Figures to exclude details such as position of components and certain precise connections between such components when such details are unnecessary for a complete understanding of how to make and use the present technology. Many of the details, dimensions, angles, and other features shown in the Figures are merely illustrative of particular embodiments of the disclosure. Accordingly, other embodiments can have other details, dimensions, angles, and features without departing from the spirit or scope of the present technology.

(31) With regard to the terms “distal” and “proximal” within this description, unless otherwise specified, the terms can reference a relative position of the portions of a catheter subsystem with reference to an operator and/or a location in the vasculature. Also, as used herein, the designations “rearward,” “forward,” “upward,” “downward,” etc., are not meant to limit the referenced component to use in a specific orientation. It will be appreciated that such designations refer to the orientation of the referenced component as illustrated in the Figures; the systems of the present technology can be used in any orientation suitable to the user.

(32) The headings provided herein are for convenience only and should not be construed as limiting the subject matter disclosed.

I. OVERVIEW

(33) FIG. 1 is a diagram of a tricuspid valve and surrounding anatomy at which a cardiac valve repair device can be implanted in accordance with embodiments of the present technology. The tricuspid valve is an atrio-ventricular valve separating the right atrium from the right ventricle, and is placed in a more apical position than the mitral valve. The tricuspid valve lies within the right trigone of the fibrous skeleton of the heart. The tricuspid annulus (a fibrous or membranous structure) constitutes the anatomical junction between the right ventricle and the right atrium, and provides a firm support structure for the tricuspid valve. The annulus is less fibrous than other annuli and slightly larger than the mitral valve annulus.

(34) The tricuspid valve has an ovoid shape and includes an anterior leaflet (also referred to as an infundibular or anterosuperior leaflet), a septal leaflet (also referred to as a medial leaflet), and a posterior leaflet (also referred to as an inferior or marginal leaflet). The anterior and septal leaflets are larger than the posterior leaflet. The fixed ends of the leaflets are attached to the annulus. The tricuspid valve has (i) a posteroseptal commissure that defines a distinct area where the septal and posterior leaflets come together at their insertion into the annulus, a (ii) an anteroseptal commissure that defines a distinct area where the septal and anterior leaflets come together at their insertion into the annulus, and (iii) an anteroposterior commissure that defines a distinct area where the anterior and posterior leaflets come together at their insertion into the annulus. The commissures can appear more like indentations than true commissures, and several millimeters of valvular tissue separate the free edges of the commissures from the annulus.

(35) The septal leaflet has more support from the fibrous trigone than the anterior or posterior leaflets. Therefore, tricuspid regurgitation from annular dilation often occurs due to the loss of

coaptation between the anterior and posterior leaflets. In addition to annular dilation, leaflet coaptation can also be adversely affected by annular calcification. The tricuspid annulus is surrounded by several important anatomic structures, including the left pulmonary artery, the coronary sinus, and the AV node. As a result, implanted cardiac devices at the tricuspid valve need to be positioned to accommodate the asymmetrical anatomy of the tricuspid valve without impacting the surrounding cardiac structures.

(36) FIGS. 2A-2D are a front view, a side view, a rear view, and a top view, respectively, of a tricuspid valve repair device **200** (“device” or “valve repair device”) that can be implanted in a heart of a subject (e.g., a human patient) in accordance with embodiments of the present technology. Referring to FIGS. 2A-2D together, the device includes a coaptation member **210**, a stabilization **220** member extending from the coaptation member **210**, and a pair of clip mechanisms **230** (also referred to as “capture clips”) movably (e.g., hinged, pivotably, rotatably) coupled to the coaptation member **210**. In some embodiments, the device **200** can include some features generally similar or identical to one or more of the implantable devices described in (i) U.S. patent application Ser. No. 16/044,447, titled “PROSTHETIC LEAFLET DEVICE,” and filed Jul. 24, 2018; (ii) International Patent Application No. PCT/US2018/061126, titled “LEAFLET EXTENSION FOR CARDIAC VALVE LEAFLET,” and filed Nov. 14, 2018; (iii) U.S. patent application Ser. No. 16/745,246, titled “IMPLANTABLE COAPTATION ASSIST DEVICES WITH SENSORS AND ASSOCIATED SYSTEMS AND METHODS,” and filed Jan. 16, 2020; (iv) U.S. patent application Ser. No. 16/817,464, titled “CARDIAC VALVE REPAIR DEVICES WITH ANNULOPLASTY FEATURES AND ASSOCIATED SYSTEMS AND METHODS,” and filed Mar. 12, 2020; and/or (v) U.S. patent application Ser. No. 17/027,681, titled “VALVE REPAIR DEVICES WITH COAPTATION STRUCTURES AND MULTIPLE LEAFLET CAPTURE CLIPS,” and filed Sep. 21, 2020; each of which is incorporated herein by reference in its entirety. Any of the valve repair devices disclosed herein can be delivered to the tricuspid valve intravascularly (e.g., trans-septal delivery via the femoral or axial vein), percutaneously (e.g., transapically), and/or surgically.

(37) In the illustrated embodiment, the coaptation member **210** is configured to (i) fill at least a portion of a regurgitant orifice between the native leaflets of the tricuspid valve, (ii) displace at least a portion of one or more of the native leaflets, and/or (iii) provide a prosthetic coaptation surface for one or more of the native leaflets. The clip mechanisms **230** are configured to be positioned on the ventricular (e.g., sub-annular) side of the tricuspid valve and to extend behind and grasp portions of one or more of the native leaflets to affix the leaflets to the coaptation member **210**. The stabilization **220** member is configured to be positioned at least partially on the atrial (e.g., supra-annular) side of the tricuspid valve and to contact the atrial sides of one or more of the native leaflets and/or other portions of the cardiac anatomy (e.g., the right atrial wall) to stabilize and secure the position of the coaptation member **210** relative to the tricuspid valve. The stabilization member **220** can also serve to inhibit or even prevent prolapse of the coaptation member **210** during ventricular systole. The stabilization member **220** can also serve to provide a platform for tissue ingrowth and long-term fixation. In the illustrated embodiment, the coaptation member **210** has a trapezoidal side-cross sectional shape and an almond-like transverse cross-sectional shape. The coaptation member **210** can further include a pair of recesses **211** (FIG. 2C) for receiving at least a portion of the clip mechanisms **230**.

(38) In the illustrated embodiment, the stabilization member **220** includes a frame **221** having an M-like shape covered by a covering **222**. In other embodiments, the frame **221** can have other shapes such as, for example, circular, elliptical, polygonal, irregular, rectilinear, and so on. The frame **221** can be a wireform, braid, or laser-cut stent-like structure formed from a suitably strong biocompatible material such as, for example, stainless steel, nickel-titanium alloys (e.g., nitinol), and/or other suitable stent materials. In some aspects of the present technology, the M-like shape of the frame **221** can provide the stabilization member with lateral stiffness (e.g., from side to side

along the stabilization member), while preserving torsional and front-to-back stability so as not to translate loads to/from the coaptation member **210**. In some embodiments, the covering **222** (e.g., fabric, graft material) can extend over at least a portion of the frame **221** to at least partially enclose the frame **221** and provide a smooth, atraumatic surface for contacting with the right atrium and/or other portions of the cardiac anatomy while promoting ingrowth into the annulus and right atrium. In some embodiments, the stabilization member **220** can have frictional elements (not shown) which engage the supra-annular and annular tissue and provide additional fixation and stability.

(39) In some embodiments, the coaptation member **210** can extend away from a downstream portion of the stabilization member **220** along a flow axis of the device **200**, and at least a portion of the coaptation member **210** can extend radially inward from the stabilization member **220** to, for example, fill a portion of the native valve orifice. In the illustrated embodiment, the stabilization member **220** is angled or biased outwardly from the coaptation member **210** by an angle Ang (FIG. 2B) of between about 10°-75° (e.g., about 15°, about 45°, more than about 45° to, for example, (i) provide stiffness and support for the coaptation member **210** and/or (ii) push a portion of an adjacent native leaflet back from the tricuspid valve opening and approximate a closed position of the native leaflet when the device **200** is implanted at the tricuspid valve. In some embodiments, the angle Ang can be selected to inhibit the coaptation member **210** from contacting the right ventricular wall during the cardiac cycle and, in particular, during systole. In some embodiments, the coaptation member **210** is more centrally located within the tricuspid valve orifice. The coaptation member **210** can be substantially stationary (e.g., little to no movement) during cardiac cycles such that the position of the coaptation member **210** relative to the stabilization member **220** is at least substantially fixed when the device **200** is deployed at the tricuspid valve. Thus, unlike native leaflets that move back and forth to open and close the native tricuspid valve, the coaptation member **210** can remain stationary during diastole and systole. In some embodiments, the coaptation member **210** does undergo some movement during cardiac cycling.

(40) An outer portion **212** of the coaptation member **210** may have a smooth, atraumatic surface (also referred to as a “coaptation surface”) for coapting with at least a portion of one or more opposing native leaflets, whereas an opposing inner portion **213** of the coaptation member **210** adjacent the clip mechanisms **230** can displace and engage at least a portion of another native leaflet. In some embodiments, the inner portion **213** and/or the outer portion **212** may include friction elements that engage the native leaflets. The coaptation member **210** can include an inner expandable frame structure (obscured in FIGS. 2A-2D; e.g., a mesh structure, a laser-cut stent frame) made from a plurality of connected struts that define an at least partially hollow interior space when the device **200** is in the illustrated deployed state. Portions of the frame structure may be disconnected allowing portions of the struts to slide over one another and/or move apart from one another to facilitate a low profile in a delivery state and/or adjustability of the coaptation member **210** dimensions. In some embodiments, the coaptation member **210** or portions thereof can be integral with the stabilization member **220**. In other embodiments, the coaptation member **210** is a separate structure that is connected to a portion of the stabilization member **220** during manufacturing using welding, rivets, adhesives, connectors, sutures and fabric, and/or other suitable connection mechanisms.

(41) The coaptation member **210** can include one or more access openings **219**, such as slits, valves, and/or holes that provide access to the interior of the coaptation member **210** and components therein during delivery and/or retrieval. For example, the access openings **219** can provide access to delivery system connectors that allow for manipulation of the coaptation member **210** and/or clip actuation mechanisms for opening and closing the clip mechanisms **230**. Further, the cavity of the coaptation member **210** may house extension members, supplemental clips, and/or other components that may be optionally deployed during implant procedures.

(42) The clip mechanisms **230** extend from the coaptation member **210** (e.g., the inner portion **213** of the coaptation member) to allow the clip mechanisms **230** to extend behind and capture one or

more native leaflets positioned on one or multiple sides of the coaptation member **210**. With reference to FIGS. **2B** and **2C**, the clip mechanisms **230** can include a base portion **231** (also referred to as a “first portion”) affixed to the coaptation member **210**, a free end portion **232** (also referred to as a “second portion”) unaffixed to the coaptation member **210**, and an articulatable arm member **233** that extends from the base portion **231** and forms the free end portion **232**. The base portion **231** can be attached to the coaptation member **210** by welding, riveting, adhesives, sutures, and/or other coupling mechanisms, or may be an extension of the coaptation member frame. The arm member **233** can extend from the base portion **231** in an upstream direction (e.g., toward the stabilization member **220**) along a length of the coaptation member **210**. For example, the arm member **233** can extend only partway up the coaptation member **210** and along the length of the coaptation member **210** to the downstream end of the stabilization member **220**. In some embodiments, the arm member **233** may form an inverted U-like shape and flare outwardly to form a wider section where the arm member **233** clamps against the native leaflet. In other embodiments, the arm member **233** may have other suitable shapes for engaging leaflets and/or may include extensions at the distal-most end that engage sub-annular tissue for additional sub-annular stabilization and fixation.

(43) The arm member **233** can be made from one or more wires, struts, and/or other semi-rigid/rigid structures with sufficient rigidity to clamp against a native leaflet and/or sub-annular tissue. In some embodiments, the arm member **233** includes a fabric covering, a biocompatible foam or other type of padding, and/or a coating on the rigid member to provide (i) a smooth surface at the arm root to reduce trauma to the leaflets and/or surrounding tissue, (ii) additional surface area for leaflet engagement, (iii) a platform for tissue ingrowth, and/or (iv) to provide additional friction to prevent leaflet slip-out. In some embodiments, the arm member **233** and/or other portions of the clip mechanism **230** can include spikes, tines, corrugations, or other frictional features (not shown) that enhance the stability and fixation to the native leaflet.

(44) The clip mechanism **230** can further include an actuation mechanism **234**, such as a spring-loaded lever, that acts on the arm member **233** to move it between a closed position (shown in FIGS. **2A-2D**; also referred to as a “closed state,” “closed configuration,” or “first state”) and an open position (also referred to as an “open state,” “open configuration,” or “second state”). In the closed state, the arm member **233** is positioned close to or against the surface of the coaptation member **210** in the corresponding recess **211**, with at least a portion of the arm member **233** pressed against the surface of the coaptation member **210** to provide for leaflet engagement. In the open state, the articulatable arm member **233** extends away from the coaptation member **210** (e.g., forming a V-shape or L-shape with the surface of the coaptation member **210**) to allow the free end portion **232** to extend behind a native leaflet and receive the native leaflet between the arm member **233** and the surface of the coaptation member **210**. In some embodiments, the actuation mechanism **234** holds the clip mechanism **230** in a normally closed state (e.g., due to a spring force) such that (i) the clip mechanism **230** is in the closed state during device delivery and (ii) manipulation of the actuation mechanism **234** moves the clip mechanism **230** to the open state. In other embodiments, the clip mechanism **230** is arranged in a normally open state.

(45) The actuation mechanism **234** can be a spring-loaded lever (e.g., a nitinol wire, laser cut nitinol or Co—Cr sheet) operably coupled to a portion of a delivery system (not shown) that can be manipulated to move the clip mechanism **230** between the open and closed positions. For example, a tendon (made of suture or nitinol wire) can be attached to the spring-loaded lever **234**, extend alongside or through the body of the coaptation member **210** and through a delivery catheter to an external handle assembly. A clinician can pull on or otherwise apply tension to the tendon, which translates this force to the lever, thereby moving the arm member **233** between the closed and open positions. In other embodiments, the actuation mechanism **234** may have different actuation means, such as other springs, clamps, pulleys, interfacing threaded members, and/or further actuation mechanisms described in International Patent Application No. PCT/US2018/061126, filed Nov. 14,

2018. Further, because each clip mechanism **230** includes its own actuation mechanism **234**, the clip mechanisms **230** can be independently actuated. As described in detail below, in some embodiments the device **200** can include more than two clip mechanisms **230** and/or one of the clip mechanisms **230** may be omitted. The actuation mechanism **234** for the clip mechanism **230** can also have a locking mechanism to prevent clip actuation after deployment.

(46) In some other embodiments, the valve repair device **200** can omit the stabilization member **220**, the stabilization member **220** can have a different shape, and/or the number and position of the clip mechanisms **230** can be varied. For example, FIGS. 3A-3D are a front view, a side view, a rear view, and a top view, respectively, of the valve repair device **200** with the stabilization member **220** omitted in accordance with embodiments of the present technology. For example, FIGS. 4A-4D are a front view, a side view, a rear view, and a top view, respectively, of the valve repair device **200** including a flexible, generally-round stabilization member **420** in accordance with embodiments of the present technology. In the illustrated embodiment, the stabilization member **420** includes a frame **421** having a number of generally-circular rings attached to the coaptation member **210** at a portion of their perimeter to, for example, provide (i) radial flexibility while minimizing lateral flexibility and (ii) a target location for additional sequential fixation to stabilize the rings against the atrial wall. And, for example, FIGS. 5A and 5B are a side view and a rear view, respectively, of a valve repair device **500** including an elongate-curved stabilization member **520** extending from a coaptation member **510** in accordance with embodiments of the present technology. In the illustrated embodiment, the stabilization member **520** includes a frame **521** having two tall wireform or laser cut structures that curve to, for example, track the shape of and brace against an atrial wall. Further, the coaptation member **510** includes only a single, centrally-located clip mechanism **530** depending therefrom.

II. SELECTED EMBODIMENTS OF TRICUSPID VALVE REPAIR DEVICES INCLUDING COAPTATION MEMBERS

(47) In general, tricuspid valve repair devices in accordance with the present technology can include a coaptation member having a shape (e.g., transverse cross-sectional shape, side cross-sectional shape, three-dimensional volumetric shape) and size (e.g., volume, area, cross-sectional dimension) selected to correspond with the natural shape of a coaptation line of the tricuspid valve to, for example, fill a leak between and/or provide a coaptation surface for one more native leaflets of the tricuspid valve. For example, all or a portion of the coaptation member can be positioned (e.g., centrally positioned) between the leaflets, or positioned with a bias toward one or more of the leaflets and/or the annulus of the tricuspid valve. The coaptation member can be positioned to primarily displace one or more of the leaflets and/or primarily to fill a commissural gap between two or more of the leaflets. The shape of the coaptation member can be selected to encourage leaflet coaptation, fill areas of regurgitation, fixate leaflets into clip mechanisms, provide a coaptation surface, facilitate native coaptation in areas not in contact with the implant device, and/or suppress native leaflet flail. For example, shapes that narrow in the coaptation zone may pull the leaflets together, increasing coaptation depth along the coaptation line and creating an annuloplasty effect. Conversely, shapes that widen along the coaptation zone can uniquely fill regurgitation spaces (e.g., clefts) in the distended tricuspid valve anatomy, further create coaptation redundancies, and/or fill space left by the native leaflets. In some embodiments, the slight annuloplasty effect of approximating the leaflets can combine with the coaptation redundancy of the coaptation member to create an overall more competent valve. In some embodiments, the size and/or orientation of the coaptation member can be adjusted by a delivery system used to deliver the tricuspid valve repair device before the delivery system is removed.

(48) In some embodiments, the coaptation member can be shaped with an atrial to ventricular gradient—such as a taper or twist—configured to direct forward flow, minimize transvalvular gradient, maximize pressure recovery, promote native leaflet closure, and/or mimic the natural eddies of blood flow throughout the cardiac cycle. The coaptation member can be covered by (i) a

fabric covering on a non-coapting face that facilitates ingrowth into the leaflets to provide robust long-term fixation, and (ii) a smooth, non-woven textile (e.g. ePTFE) covering on a coapting face that provides an atraumatic surface for coapting with the native leaflets. In some embodiments, the coaptation member can include foam under the fabric covering to provide for further atraumatic coaptation of leaflets against the coaptation member. The coaptation member can be supported by braided wire, superelastic nitinol stent-like frames, expanding sponge-like materials, polymer balloons, and/or other support structures.

(49) More specifically, for example, FIGS. **6A-6H** are transverse cross-sectional views (e.g., top or atrial views, bottom or ventricular views) of various coaptation members **610** in accordance with embodiments of the present technology. As shown in FIGS. **6A-6H**, respectively, the coaptation member **610** can have a crescent shape, oval shape, elongated polygonal shape, triangular or asymmetrical hexagonal shape, semicircular shape, mushroom-like or umbrella-like shape, T-shape, and/or star shape (e.g., having three or more points). Similarly, FIGS. **7A-7L** are side cross-sectional views (e.g., anterior-posterior views and/or commissure-commissure views) of various coaptation members **710** in accordance with embodiments of the present technology. As shown in FIGS. **7A-7L**, respectively, the coaptation member **710** can have a triangular shape (e.g., isosceles triangular shape), curved or fin-like shape, oval shape, inverted-triangular shape, elongated T-shape, inverted umbrella-like shape, trapezoidal shape, semicircular shape, laterally-elongated T-shape, square shape, circular shape, or bow-tie-like shape (e.g., including a pair of trapezoidal portions extending from a central member). The various transverse cross-sectional and side cross-sectional shapes of the coaptation members **610** and **710** can be combined to form coaptation members of different shapes and sizes. Likewise, in other embodiments, coaptation members in accordance with the present technology can have other shapes.

(50) The various coaptation member shapes illustrated in FIGS. **6A-7L** can be combined and/or modified based on, for example, the particular anatomy and/or abnormality of the tricuspid valve at which the valve repair device is to be implanted. For example, FIG. **8A** is a side cross-sectional view of a valve repair device **800** implanted at a tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **8B** and **8C** are transverse cross-sectional views of the valve repair device **800** of FIG. **8A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **8A-8C** together, the valve repair device **800** includes a coaptation member **810** having a generally circular side cross-sectional shape and an oval transverse cross-sectional shape. In the illustrated embodiment, the coaptation member **810** can be secured to and/or against a first leaflet **L1** of the tricuspid valve TV, such as the posterior leaflet, via one or more clip mechanisms, anchors, and/or other securing features described herein (not shown). In other embodiments, the coaptation member **810** can additionally or alternatively be secured to and/or against a second leaflet **L2** (e.g., the anterior leaflet) and/or a third leaflet **L3** (e.g., the septal leaflet). The coaptation member **810** provides a coaptation surface for the second and third leaflets **L2** and **L3** as shown in, for example, FIG. **8C**.

(51) FIG. **9A** is a side cross-sectional view of a valve repair device **900** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **9B** and **9C** are transverse cross-sectional views of the valve repair device of FIG. **9A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **9A-9C** together, the valve repair device **900** includes a coaptation member **910** having a generally pentagonal side cross-sectional shape and a three-pointed-star-like transverse cross-sectional shape. In the illustrated embodiment, the valve repair device **900** can be secured between the leaflets **L1-L3** of the tricuspid valve TV via one or more clip mechanisms, lock mechanisms, anchors, and/or other securing features described herein (not shown). As best seen in FIGS. **9B** and **9C**, the star-like transverse cross-sectional shape of the coaptation member **910** can be oriented such that each of the points of the star generally points toward a corresponding one of the commissures between the leaflets **L1-L3** to, for example, facilitate the coaptation of the leaflets **L1-L3** against the surface of

the coaptation member **910**. In other embodiments, the coaptation member **910** of the valve repair device **900** can have any of the side cross-sectional shapes shown in FIG. 9D including, for example, a curved-pentagonal, squared-semicircular, or rectangular shape.

(52) FIG. **10A** is a top view of a valve repair device **1000** in accordance with embodiments of the present technology. FIG. **10B** is a side cross-sectional view of the valve repair device **1000** of FIG. **10A** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **10C** and **10D** are transverse cross-sectional views of the valve repair device of FIGS. **10A** and **10B** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **10A-10D** together, the valve repair device **1000** includes a coaptation member **1010** with generally triangular (e.g., teardrop-shaped, ovoid-shaped) side and transverse cross-sectional shapes. The valve repair device **1000** further includes a fixation mechanism **1035** extending from one or more edges of coaptation member **1010** and configured to secure the coaptation member **1010** to two or more of the leaflets of the tricuspid valve, such as the leaflets L1 and L2 of the tricuspid valve TV. In other embodiments, the fixation mechanism **1035** can include one or more clip mechanisms, lock mechanisms, anchors, and/or other securing features described herein for fixating the device onto the leaflets of the valve. Accordingly, the coaptation member **1010** can (i) be positioned in the commissure between the leaflets, (ii) extend into the center of the tricuspid valve TV, and (iii) be shaped to fill a regurgitant orifice and prevent regurgitation. In some embodiments, the coaptation member **1010** can be biased toward an annulus A of the tricuspid valve TV to help fill a regurgitant space in the valve.

III. SELECTED EMBODIMENTS OF TRICUSPID VALVE REPAIR DEVICES INCLUDING MECHANISMS FOR SECURING TO CARDIAC ANATOMY GENERALLY LOCAL TO THE TRICUSPID VALVE

(53) In some embodiments, cardiac valve repair devices in accordance with the present technology can include implant fixation mechanisms for securing the device into the local native anatomy of the tricuspid valve by, for example, anchoring into or laying against the native ventricular wall, against the annulus, and/or onto the leaflets themselves. In some embodiments, the implant fixation mechanisms can include one or more clip mechanisms and one or more lock mechanisms. The clip mechanisms (also referred to as “clips,” “capture clips,” “capture mechanisms,” and iterations thereof) are configured to be positioned on the ventricular side of the tricuspid valve and to capture one or more leaflets of the valve for securing the leaflets against, for example, a coaptation member coupled to the clip mechanisms and lock mechanisms. The lock mechanisms (also referred to as “clips,” “locking clips,” “stabilization members,” “stabilization features”, and iterations thereof) can be generally similar to the clip mechanisms but are configured to be positioned on the atrial side of the tricuspid valve and to engage the atrial side of the valve leaflets and/or other portions of the cardiac anatomy to, for example, help secure the coaptation member in a selected position relative to the tricuspid valve and/or provide additional leaflet fixation. The clip and lock mechanisms may or may not require a specific orientation and can be interchanged in functionality.

(54) In some embodiments, the clip mechanisms can be on (i) opposing sides of the coaptation member, (ii) one side of the coaptation member, and/or (iii) a mating surface inferior to the coaptation member. In some embodiments, the clip mechanisms and/or the lock mechanisms can be movable, expandable, and/or otherwise adjustable. In some embodiments, the clip mechanisms can be narrow numerate features configured to navigate chordae proximate to the tricuspid valve and/or to minimize leaflet disruption during diastole (e.g., allowing forward flow), and the lock mechanisms can be wide features on the atrial side to minimize leaflet flail and provide additional flow resistance during systole. In some embodiments, the clip and lock mechanisms can operate independently from one another and can be repositionable. Further, the clip and lock mechanisms (e.g., arms thereof) can (i) have various shapes supporting the native leaflet shapes, (ii) be set at an angle to mimic the natural leaflet coaptation angle, and/or (iii) be configured to capture only the free edge of one or more of the leaflets.

(55) In some embodiments, the implant fixation mechanisms can include features configured to enhance leaflet fixation, such as (i) interlocking components configured to increase leaflet plication, (ii) materials or features that increase surface area in contact with the leaflets, (iii) frictional elements (e.g., cleats, barbs, textures) that increase friction against the leaflets, (iv) features that puncture the leaflets, and/or (v) combinations thereof. For tricuspid valve repair devices having clips on the atrial and ventricular sides of the device, features on the atrial side of the leaflet (e.g., lock mechanisms) may fit within the spacing of the sub-valvular features. Fixation features can be gear driven, hydraulic, superelastic, or spring-loaded. Features that contact the leaflet can be widened to distribute the closing force across a wider area. Some stabilizing features can utilize flared, angled, or wide fixation mechanisms to add stability to the device and facilitate the natural coaptation angle of the native valve. In some embodiments, the clip mechanisms and/or the lock mechanisms can include a patent foramen ovale (PFO) closure device configured as an atrial anchor.

(56) In some embodiments, the clip mechanisms and/or the lock mechanisms can be in the form of a hook so as to not tightly pull the leaflet up against the coaptation member but to just approximate it. This configuration can allow for a captured leaflet to open more during diastole thereby reducing pressure gradients. When multiple (e.g., two) ventricular clip mechanisms are in the form of hooks, atrial annular support and/or anchor members can be included to inhibit migration of the device into the ventricle. In some embodiments, the atrial anchors can be inserted into the septal wall through a stabilizing member connected to the coaptation member. In some embodiments, the clip mechanisms can include one or more clip expandable arms that work efficiently in treating a wide mitral regurgitation jet without the need for a coaptation member with a C—C protrusion.

(57) FIG. 11A is a side cross-sectional view of a valve repair device **1100** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. 11B and 11C are transverse cross-sectional views of the valve repair device **1100** of FIG. 11A during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. 11A-11C together, the valve repair device includes features generally similar to the valve repair device **1000** of FIGS. 10A-10D including, for example, a coaptation member **1110** secured to the leaflets L1 and L2. In the illustrated embodiment, however, the coaptation member **1110** has a fixation mechanism including (i) clip mechanisms **1130** configured to engage the ventricular sides of the leaflets L1 and L2 and (ii) lock mechanisms **1136** configured to engage the atrial side of the leaflets L1 and L2. In addition, the coaptation member **1110** orientation has been reversed from the valve repair devices **1000** of FIGS. 10A-10D with the narrow portion of the surface of the coaptation member **1110** oriented towards the commissure between the leaflets. Accordingly, in some aspects of the present technology the coaptation member **1110** can be oriented and/or placed in versatile manner between the leaflets. In general, the various coaptation members of the present technology can be optionally placed in different orientations depending on the specific application of the valve and valve repair device.

(58) FIG. 12A is a side cross-sectional view of a valve repair device **1200** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. 12B and 12C are transverse cross-sectional views of the valve repair device of FIG. 12A during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. 12A-12C together, the valve repair device **1200** includes a coaptation member **1210** with a generally elongated-triangular shape (e.g., a “bicycle-seat” shape). The coaptation member **1210** can further include (i) a supra-valvular portion **1214** secured (e.g., anchored) to the annulus A and/or one or more of the leaflets of the tricuspid valve TV via a first fixation mechanism **1235a** and (ii) a sub-valvular portion **1215** anchored to one or more of the leaflets of the tricuspid valve TV via second fixation mechanism (not shown). In some embodiments, the first fixation mechanism **1235a** can be an anchor, such as a helical screw, and the second fixation mechanism can be a clip mechanism. Accordingly, the coaptation member **1210** can replace all or part of a native leaflet of the tricuspid

valve TV (e.g., the leaflet L1) while providing a coaptation surface for one or more of the other native leaflets (e.g., the leaflets L2 and L3). In other embodiments, the second fixation mechanism can additionally or alternatively include an anchor or hook feature configured to secure the sub-valvular portion **1215** of the coaptation member **1210** to a portion of the cardiac anatomy at or near a right ventricular outflow tract RVOT.

(59) In yet other embodiments, the second fixation mechanism can be an anchor (e.g., a second helical screw) configured to secure the coaptation member to the annulus A of the tricuspid valve TV. For example, FIG. **13A** is a side cross-sectional view of a valve repair device **1300** similar to FIGS. **12A-12C** implanted at the tricuspid valve TV with a first fixation mechanism **1335a** and a second fixation mechanism **1335b** anchoring the coaptation member **1310** to the annulus A in accordance with embodiments of the present technology. FIGS. **13B** and **13C** are transverse cross-sectional views of the valve repair device **1300** of FIG. **13A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **13A-13C**, the second fixation mechanism **1335b** can be secured to the annulus A such that most or all of the coaptation member **1310** is positioned at the coaptation surface of the tricuspid valve TV.

(60) FIG. **14A** is a front view of a tricuspid valve repair device **1400** in accordance with embodiments of the present technology. FIG. **14B** is a side cross-sectional view of the valve repair device **1400** of FIG. **14A** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **14C** and **14D** are transverse cross-sectional views of the valve repair device **1400** of FIGS. **14A** and **14B** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **14A-14D** together, the valve repair device **1400** can include features generally similar or identical to those of the tricuspid valve repair device **200** of FIGS. **2A-2D** including, for example, (i) a coaptation member **1410** having a pair of clip mechanisms **1430** coupled thereto and (ii) a stabilization member **1420** (also referred to as an atrial support member) extending upward away from the coaptation member **1410**. The stabilization member **1420** is configured to anchor or brace against the septal wall above the tricuspid valve TV, as best seen in FIG. **14B**, to help secure the position of the coaptation member **1410** between the native leaflets. The stabilization member **1420** can also provide a platform for tissue ingrowth and long-term fixation. Additionally, the stabilization member **1420** can serve as platform to locate supplemental anchors into the atrial septum. The clip mechanisms **1430** can secure (e.g., fix) the coaptation member **1410** to one of the native leaflets (e.g., the leaflet L1; the septal leaflet) such that the coaptation member **1410** replaces all or part the leaflet L1 and provides a coaptation surface for one or more of the other native leaflets.

(61) In some embodiments, the stabilization member **1420** is configured to puncture the septal wall to further stabilize the coaptation member **1410** at the tricuspid valve TV. For example, FIG. **14E** is a front view of the valve repair device **1400** of FIGS. **14A-14D** including a puncture feature **1424** on the stabilization member **1420** in accordance with embodiments of the present technology. The puncture feature **1424** is configured to puncture the septal wall to fixate the stabilization member **1420** thereto.

(62) FIG. **15A** is a top view of a valve repair device **1500** in accordance with embodiments of the present technology. FIG. **15B** is a side cross-sectional view of the valve repair device **1500** of FIG. **15A** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. Referring to FIGS. **15A** and **15B** together, the valve repair device **1500** includes a coaptation member **1510** (also referred to as a “baffle”), a stabilization member **1520** (also referred to as a “chair-back”) extending from the coaptation member **1510**, and a clip mechanism **1530** extending from the coaptation member **1510**. The stabilization member **1520** can engage an atrial or septal wall above the tricuspid valve TV to help stabilize the position of the coaptation member **1510** between one or more native leaflets of the tricuspid valve TV. The clip mechanism **1530** can engage the leaflet L1 to secure the coaptation member **1510** against the leaflet L1. In some embodiments, the stabilization member **1520** is positioned against the septal wall between the right

atrium and the left atrium and the leaflet L1 is the septal leaflet of the tricuspid valve TV.

(63) In the illustrated embodiment, the stabilization member **1520** is spaced apart from the coaptation member **1510** by a gap **1525** (FIG. **15A**) to, for example, enable the coaptation member **1510** to be deployed from a delivery system and tested before deploying the stabilization member **1520** from the delivery system. The stabilization member **1520** may be connected to the coaptation member **1510** through sutures and fabric, small superelastic wires, and/or other attachment mechanisms to, for example, maintain independence and flexibility while providing stabilization. In some embodiments, a top portion **1516** of the coaptation member **1510** and a top portion **1526** of the stabilization member **1520** can each have a triangular, angled, or “chevron-like” shape to, for example, enable the valve repair device **1500** to be more easily recaptured or recovered into the delivery system.

(64) FIG. **16A** is a side cross-sectional view of a valve repair device **1600** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **16B** and **16C** are transverse cross-sectional views of the valve repair device **1600** of FIG. **16A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **16A-16C** together, the valve repair device **1600** includes a coaptation member **1610** and a plurality of clip mechanisms **1630** configured to secure the coaptation member **1610** to two or more of the native leaflets of the tricuspid valve TV. In the illustrated embodiment, for example, two of the clip mechanisms **1630** fix the coaptation member to the leaflet L1 (e.g., the septal leaflet) and one of the clip mechanisms **1630** fixes the coaptation member to the leaflet L2 (e.g., the anterior or posterior leaflet) to allow the remaining leaflet L3 (e.g., the other of the anterior or posterior leaflet) to coapt freely against the surface of the coaptation member **1610**. In some embodiments, the coaptation member **1610** can be shaped to fill the resultant gap between the leaflets L1-L3. For example, the coaptation member **1610** can have both a generally rounded-triangular transverse cross-sectional shape and side cross-sectional shape as shown in FIGS. **16A-16C**.

(65) In other embodiments, a coaptation member of a valve repair device can have other shapes and/or the number and/or positioning of the clip mechanisms can vary. For example, FIGS. **17A**, **18A**, and **19A** are side cross-sectional view of valve repair devices **1700**, **1800**, and **1900**, respectively, implanted at the tricuspid valve TV in accordance with additional embodiments of the present technology. FIGS. **17B**, **18B**, and **19B** are transverse cross-sectional views of the valve repair devices **1700**, **1800**, and **1900** of FIGS. **17A**, **18A**, and **19A**, respectively, during diastole in accordance with embodiments of the present technology. And, FIGS. **17C**, **18C**, and **19C** are transverse cross-sectional views of the valve repair devices **1700**, **1800**, and **1900** of FIGS. **17A**, **18A**, and **19A**, respectively, during systole in accordance with embodiments of the present technology. Referring first to FIGS. **17A-17C** together, the valve repair device **1700** can include (i) a coaptation member **1710** having a generally rounded-triangular or rounded-trapezoidal side cross-sectional shape and an almond or oval transverse cross-sectional shape, (ii) a first clip mechanism **1730a** configured to fix the coaptation member **1710** to the leaflet L1, and (ii) a second clip mechanism **1730b** configured to fix the coaptation member **1710** to the leaflet L2. Referring next to FIGS. **18A-18C** together, the valve repair device **1800** can include (i) a coaptation member **1810** having a generally triangular side cross-sectional shape and an almond or oval transverse cross-sectional shape and (ii) a pair of first clip mechanisms **1830a** configured to secure the coaptation member **1810** to the leaflet L1 and a single second clip mechanism **1830b** configured to secure the coaptation member **1810** to the leaflet L2. Referring next to FIGS. **19A-19C** together, the valve repair device **1900** can include (i) a coaptation member **1910** having a rectangular side cross-sectional shape and a bicycle-seat-like transverse cross-sectional shape and (ii) a pair of clip mechanisms **1930** configured to secure the coaptation member **1910** to the leaflet L1. Referring to FIGS. **17A-19C** together, the coaptation members **1710**, **1810**, and/or **1910** may be shaped per any of the shapes illustrated in FIGS. **6A-6H** and to extend through the tricuspid valve TV such that at least a portion of the coaptation member has a sub-valvular position in order to provide a

coaptation surface across a range of valve etiologies and redundancies.

IV. SELECTED EMBODIMENTS OF TRICUSPID VALVE REPAIR DEVICES INCLUDING MECHANISMS FOR SECURING TO CARDIAC ANATOMY GENERALLY REMOTE FROM THE TRICUSPID VALVE

(66) In some embodiments, a tricuspid valve repair device in accordance with embodiments of the present technology can include (i) a coaptation member positioned between the native valve leaflets, and (ii) one or more anchors, tethers, and/or support members that additionally or alternatively secure the coaptation member to anatomy of the heart other than the native leaflets, annulus, or other local anatomy of the tricuspid valve. For example, the anchors can be secured to the right atrial wall, right ventricular wall, right ventricular outflow tract, inferior vena cava, superior vena cava, and/or other portions of the anatomy of the right heart of the patient.

(67) FIG. 20A is a side cross-sectional view of a valve repair device **2000** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. 20B and 20C are transverse cross-sectional views of the valve repair device **2000** of FIG. 20A during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. 20A-20C together, the valve repair device **2000** includes a coaptation member **2010** and a pair of stabilization members **2040** (which can also be referred to as “anchoring mechanisms,” “extensions,” and the like) extending from the coaptation member **2010** into the right ventricular outflow tract RVOT. The stabilization members **2040** can be anchored (e.g., fixed, attached) within the right ventricular outflow tract RVOT, or can be unanchored and merely press against anatomy thereof to secure the coaptation member **2010** in position relative to the native leaflets of the tricuspid valve TV. In some embodiments, the coaptation member **2010** and the operation of the valve repair device **2000** can be generally similar or identical to that of the tricuspid valve repair devices **200** and/or **1400** described in detail above with reference to FIGS. 2A-2D and 14A-14E. For example, the stabilization members **2040** can secure the coaptation member **2010** to one of the native leaflets (e.g., the leaflet L1) such that the coaptation member **2010** replaces all or part of the native leaflet while providing a coaptation surface for one or more of the other native leaflets (e.g., the leaflets L2 and L3).

(68) In some embodiments, the valve repair device **2000** can further include an atrial stabilization member. For example, FIG. 24A is a side cross-sectional view of the valve repair device **2000** of FIGS. 23A-23C further including an atrial stabilization member **2120** and implanted at the tricuspid valve TV in accordance with additional embodiments of the present technology. FIGS. 24B and 24C are transverse cross-sectional views of the valve repair **2000** device of FIG. 24A during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. 24A-24C together, the atrial stabilization member **2120** is configured to anchor or brace against the septal wall above the tricuspid valve TV, as best seen in FIG. 24A, to help secure the position of the coaptation member **2010** between the native leaflets together with the ventricular stabilization members **2040** that reside within the right ventricular outflow tract RVOT.

(69) FIG. 22 is a side cross-sectional view of a valve repair device **2200** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. In the illustrated embodiment, the valve repair device **2200** includes a coaptation member **2210** fixed to one or more of the native leaflets (e.g., the septal leaflet) of the tricuspid valve TV via a clip mechanism **2230**. The valve repair device **2200** can further include an anchor **2242** secured in the right ventricular outflow tract RVOT, and a tether **2243** (also referred to as a “support element”, “connector,” and the like) extending between the coaptation member **2210** and the anchor **2242**. In some embodiments, the anchor **2242** can include a laser-cut stent, braded stent, mesh and/or other element that imparts a radially outward force against the right ventricular outflow tract RVOT to secure (e.g., fix) the anchor **2242** in position. In some embodiments, the anchor **2242** can include a braided porous mesh (e.g., a filter) configured to permit blood flow into the pulmonary artery. In

some embodiments, the anchor **2242** can include atraumatic frictional elements (not shown) on an outer surface thereof and configured to further fix the anchor **2242** in the right ventricular outflow tract RVOT. The tether **2243** can be a metallic wire, metallic suture, polymer suture, and/or another elongate element connected between the coaptation member **2210** and the anchor **2242**.

(70) In other embodiments, the coaptation member **2210** can be secured against one or more other native leaflets of the tricuspid valve TV. For example, FIG. **23** is a side cross-sectional view of the valve repair device **2200** of FIG. **22** implanted at the tricuspid valve TV and secured to the anterior and/or posterior leaflets in accordance with embodiments of the present technology. For example, the clip mechanism **2230** can be secured to a different one or more of the leaflets.

(71) In some embodiments, a valve repair device in accordance with the present technology can further include an atrial stabilization member (also referred to as an “atrial fixation member”) extending from the coaptation member into the right atrium and anchored to and/or secured against the septal wall or fossa. Such atrial stabilization members can include a braided mesh (e.g., an Amplatzer-style mesh or filter), a helical screw, and/or other support structures configured to anchor into the septal wall. The atrial stabilization member can inhibit or even prevent the coaptation from migrating into the right atrium, thereby inhibiting further prolapse of the native leaflets of the tricuspid valve TV. In some embodiments, the atrial stabilization member can include radiopaque or echogenic features that facilitate placement of the stabilization member (e.g., screws thereof) into the septal wall. In some embodiments, such an atrial stabilization member can be modularly added during a delivery procedure. For example, FIGS. **24A** and **24B** are side cross-sectional views of a valve repair device **2400** implanted at the tricuspid valve TV in a first position and a second position, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **24A** and **24B** together, the valve repair device **2400** includes a coaptation member **2410**, one or more stabilization members **2440**, and an atrial stabilization member **2420**. Referring to FIG. **24A**, the valve repair device **2400** can first be deployed at the tricuspid valve TV with the atrial stabilization member **2420** in an undeployed state (e.g., not deployed into the septal wall or fossa). Referring next to FIG. **24B**, the atrial stabilization member **2420** can be fixed to the septal wall or fossa to further secure the coaptation member **2410** at the tricuspid valve TV if, for example, testing determines that further stabilization is desired.

(72) FIG. **25A** is a side cross-sectional view of a valve repair device **2500** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **25B** and **25C** are transverse cross-sectional views of the valve repair device **2500** of FIG. **25A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **25A-25C** together, the valve repair device **2500** includes a coaptation member **2510** fixed to one or more of the native leaflets of the tricuspid valve TV via a clip mechanism **2530**. The valve repair device **2500** can further include an anchor **2542** secured in the inferior vena cava IVC, and a tether **2543** extending between the coaptation member **2510** and the anchor **2542**. In some embodiments, the anchor **2542** can include a laser-cut stent, braded stent, mesh and/or other element that imparts a radially outward force against the inferior vena cava IVC to secure (e.g., fix) the anchor **2542** in position. The tether **2543** can be a metallic wire, metallic suture, polymer suture, and/or other elongate element connected to the coaptation member **2510** at an anchor connection point and configured to inhibit or even prevent atrial migration of the coaptation member **2510**.

(73) FIG. **26** is a side cross-sectional view of a valve repair device **2600** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. In the illustrated embodiment, the valve repair device **2600** includes a coaptation member **2610** fixed to one or more of the native leaflets of the tricuspid valve TV via a clip mechanism **2630**. The valve repair device **2600** can further include an anchor **2642** secured in the superior vena cava SVC, and a tether **2643** extending between the coaptation member **2610** and the anchor **2642**. In some embodiments, the anchor **2642** can include a laser-cut stent, braded stent, mesh and/or other element that imparts a

radially outward force against superior vena cava SVC to secure (e.g., fix) the anchor **2642** in position. The tether **2643** can be a metallic wire, metallic suture, polymer suture, and/or other stiff support element connected to the coaptation member **2610** and configured to inhibit or even prevent prolapse of the coaptation member **2610** into the right atrium RA.

(74) Referring to FIGS. **20A-26** together, tricuspid valve repair devices in accordance with the present technology can include various combinations of the described and illustrated anchoring mechanisms, clip mechanisms, and/or support mechanisms to, for example, provide a specific support structure for a coaptation member based on the particular anatomy of a patient. For example, a clip mechanism can provide primary fixation for the coaptation member, while a ventricular anchor to the right ventricular outflow tract RVOT can be added modularly in flail conditions to inhibit or even prevent excessive atrial movement of the coaptation member from leaflet flail. Likewise, an atrial-side anchor to the superior vena cava SVC and/or the inferior vena cava IVC can be added to inhibit or even prevent atrial migration of the coaptation member.

(75) More specifically, for example, FIG. **27A** is a side cross-sectional view of a valve repair device **2700** implanted at the tricuspid valve TV in accordance with embodiments of the present technology. FIGS. **27B** and **27C** are transverse cross-sectional views of the valve repair device **2700** of FIG. **27A** during diastole and systole, respectively, in accordance with embodiments of the present technology. Referring to FIGS. **27A-27C** together, the valve repair device **2700** includes a coaptation member **2710** secured at the tricuspid valve TV between the native leaflets via a first tether **2743a** extending to an anchor **2740** in the right ventricular outflow tract RVOT—without a clip mechanism fixing the coaptation member **2710** to any of the native leaflets. The valve repair device **2700** can optionally include anchors **2742** in the inferior vena cava IVC and/or the superior vena cava SVC and connected to the coaptation member **2710** via second tethers **2743b**. As shown in FIGS. **27B** and **27C**, in some embodiments the coaptation member **2710** can be adjustable (e.g., expandable) between a first (e.g., radially-compressed) position **2710'** and a second (e.g., radially-expanded) position **2710''**.

V. ADDITIONAL EXAMPLES

(76) The following examples are illustrative of several embodiments of the present technology:

(77) 1. A tricuspid valve repair device, comprising: a coaptation member configured to be positioned at least partially between native leaflets of a tricuspid valve; a clip mechanism coupled to the coaptation mechanism, wherein the clip mechanism is configured to engage a ventricular side of one of the native leaflets to secure the one of the native leaflets between the clip mechanism and the coaptation member, and wherein the coaptation member includes a surface configured to coapt with at least another one of the native leaflets; and a stabilization member extending from the coaptation member and configured to be positioned at least partially within a right atrium above the tricuspid valve, wherein the stabilization member is configured to contact a wall within the right atrium to stabilize a position of the coaptation member relative to the tricuspid valve.

(78) 2. The tricuspid valve repair device of example 1 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve.

(79) 3. The tricuspid valve repair device of example 1 or example 2 wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

(80) 4. The tricuspid valve repair device of any one of examples 1-3 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve, and wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

(81) 5. The tricuspid valve repair device of any one of examples 1-4 wherein the stabilization member includes a frame and a covering over the frame.

(82) 6. The tricuspid valve repair device of example 5 wherein the frame is formed from a metal material, and wherein the covering is formed from at least one of a fabric material, an extruded polymeric material, and a graft material.

(83) 7. The tricuspid valve repair device of example 5 or example 6 wherein the frame has an M-

like shape.

(84) 8. The tricuspid valve repair device of example 5 or example 6 wherein the frame includes two or more wireform loops extending from the coaptation member.

(85) 9. The tricuspid valve repair device of any one of examples 1-8 wherein the coaptation member extends radially inward from the stabilization member toward a flow axis of the tricuspid valve, and wherein the stabilization member extends from the coaptation member at an angle relative to the flow axis.

(86) 10. The tricuspid valve repair device of any one of examples 1-9 wherein the clip mechanism is one of a plurality of clip mechanisms, wherein each of the clip mechanisms is configured to engage the ventricular side of the one of the native leaflets to secure the one of the native leaflets between the clip mechanism and the coaptation member.

(87) 11. The tricuspid valve repair device of any one of examples 1-10 wherein the stabilization member includes an upper portion with a chevron-like shape.

(88) 12. The tricuspid valve repair device of example 11 wherein the coaptation member includes an upper portion with a chevron-like shape.

(89) 13. The tricuspid valve repair device of any one of examples 1-12, further comprising a puncture member coupled to the stabilization member and configured to puncture the wall within the right atrium to further stabilize the position of the coaptation member relative to the tricuspid valve.

(90) 14. A tricuspid valve repair device, comprising: a coaptation member configured to be positioned at least partially between a native septal leaflet, a native posterior leaflet, and a native anterior leaflet of a tricuspid valve, wherein the coaptation member includes a first surface and a second surface; multiple clip mechanisms coupled to the coaptation mechanism, wherein each of the clip mechanisms is configured to engage a ventricular side of the native septal leaflet to secure the native septal leaflet between the clip mechanism and the first surface of the coaptation member, and wherein the at second surface of the coaptation member is configured to coapt with at least one of the native anterior leaflet and the native posterior leaflet; and a stabilization member extending from the coaptation member and configured to be positioned at least partially within a right atrium above the tricuspid valve, wherein the stabilization member is configured to contact a septal between the right atrium and a left atrium to (a) stabilize a position of the coaptation member relative to the tricuspid valve and (b) provide a platform for tissue ingrowth and long-term fixation.

(91) 15. The tricuspid valve repair device of example 14 wherein the tricuspid valve includes a flow axis, and wherein the stabilization member extends away from the coaptation member toward the septal wall at an angle relative to the flow axis.

(92) 16. The tricuspid valve repair device of example 14 or example 15 wherein the stabilization member includes a wireform extending from the coaptation member and a covering over the wireform.

(93) 17. A method of repairing a tricuspid valve, the method comprising: deploying a coaptation member of a tricuspid valve repair device at least partially between native leaflets of a tricuspid valve; engaging a ventricular side of one of the native leaflets with a clip mechanism of the tricuspid valve repair device to secure the one of the native leaflets between the clip mechanism and a first surface of the coaptation member; positioning the coaptation member such that a second surface of the coaptation member is configured to coapt with at least another one of the native leaflets; and contacting a stabilization member of the tricuspid valve repair device with a wall within a right atrium above the tricuspid valve to stabilize a position of the coaptation member relative to the tricuspid valve.

(94) 18. The method of example 17 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve.

(95) 19. The method of example 17 or example 18 wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

(96) 20. The method of any one of examples 17-19 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve, and wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

VI. CONCLUSION

(97) The above detailed description of embodiments of the technology are not intended to be exhaustive or to limit the technology to the precise form disclosed above. Although specific embodiments of, and examples for, the technology are described above for illustrative purposes, various equivalent modifications are possible within the scope of the technology as those skilled in the relevant art will recognize. For example, although steps are presented in a given order, alternative embodiments can perform steps in a different order. The various embodiments described herein can also be combined to provide further embodiments.

(98) From the foregoing, it will be appreciated that specific embodiments of the technology have been described herein for purposes of illustration, but well-known structures and functions have not been shown or described in detail to avoid unnecessarily obscuring the description of the embodiments of the technology. Where the context permits, singular or plural terms can also include the plural or singular term, respectively.

(99) Moreover, unless the word “or” is expressly limited to mean only a single item exclusive from the other items in reference to a list of two or more items, then the use of “or” in such a list is to be interpreted as including (a) any single item in the list, (b) all of the items in the list, or (c) any combination of the items in the list. Additionally, the term “comprising” is used throughout to mean including at least the recited feature(s) such that any greater number of the same feature and/or additional types of other features are not precluded. It will also be appreciated that specific embodiments have been described herein for purposes of illustration, but that various modifications can be made without deviating from the technology. Further, while advantages associated with some embodiments of the technology have been described in the context of those embodiments, other embodiments can also exhibit such advantages, and not all embodiments need necessarily exhibit such advantages to fall within the scope of the technology. Accordingly, the disclosure and associated technology can encompass other embodiments not expressly shown or described herein. so on.

Claims

1. A tricuspid valve repair device, comprising: a coaptation member configured to be positioned at least partially between a first native leaflet and a second native leaflet of a tricuspid valve; a plurality of clip mechanisms coupled to the coaptation member, wherein each of the clip mechanisms is configured to clamp against a ventricular side of the first native leaflet to secure the first native leaflet between the clip mechanism and the coaptation member, and wherein the coaptation member includes a surface configured to coapt with at least the second native leaflet; and a stabilization member extending from the coaptation member and configured to be positioned at least partially within a right atrium above the tricuspid valve, wherein the stabilization member is shaped to extend along and press against an atrial side of the first native leaflet and a wall within the right atrium above the first native leaflet to stabilize a position of the coaptation member relative to the tricuspid valve, and wherein the stabilization member is biased outwardly from the coaptation member to push a portion of the first native leaflet back from an opening of the tricuspid valve.

2. The tricuspid valve repair device of claim 1 wherein the first native leaflet is a native septal leaflet of the tricuspid valve.

3. The tricuspid valve repair device of claim 1 wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

4. The tricuspid valve repair device of claim 1 wherein the first native leaflet is a native septal

leaflet of the tricuspid valve, and wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

5. The tricuspid valve repair device of claim 1 wherein the stabilization member includes a frame and a covering over the frame.

6. The tricuspid valve repair device of claim 5 wherein the frame is formed from a metal material, and wherein the covering is formed from at least one of a fabric material, an extruded polymeric material, and a graft material.

7. The tricuspid valve repair device of claim 5 wherein the frame has an M-like shape.

8. The tricuspid valve repair device of claim 5 wherein the frame includes two or more wireform loops extending from the coaptation member.

9. The tricuspid valve repair device of claim 1 wherein the coaptation member extends radially inward from the stabilization member toward a flow axis of the tricuspid valve, and wherein the stabilization member extends from the coaptation member at an angle relative to the flow axis.

10. The tricuspid valve repair device of claim 1 wherein the stabilization member includes an upper portion with a chevron-like shape.

11. The tricuspid valve repair device of claim 10 wherein the coaptation member includes an upper portion with a chevron-like shape.

12. The tricuspid valve repair device of claim 1, further comprising a puncture member coupled to the stabilization member and configured to puncture the wall within the right atrium to further stabilize the position of the coaptation member relative to the tricuspid valve.

13. The tricuspid valve repair device of claim 1 wherein the stabilization member is shaped to extend along and press against the atrial side of the first native leaflet and the wall within the right atrium above the first native leaflet to substantially maintain the coaptation member in a substantially stationary position relative to the tricuspid valve during cardiac cycles.

14. The tricuspid valve repair device of claim 1 wherein the stabilization member comprises a frame made of stent material that provides lateral stiffness and torsional and front-to-back stability to reduce loads transferred to and/or from the coaptation member.

15. The tricuspid valve repair device of claim 1 wherein the stabilization member comprises a frame shaped to track a shape of and brace against the wall within the right atrium above a native valve annulus.

16. The tricuspid valve repair device of claim 1 wherein the stabilization member comprises a frame with an elongate shape extending away from the coaptation member and configured to be positioned above a native valve annulus.

17. The tricuspid valve repair device of claim 1 wherein the stabilization member comprises a frame configured to stabilize the position of the coaptation member relative to the tricuspid valve without penetrating cardiac tissue.

18. The tricuspid valve repair device of claim 1 wherein the stabilization member is angled from the coaptation member by an angle between 10-75 degrees.

19. A tricuspid valve repair device, comprising: a coaptation member configured to be positioned at least partially between a native septal leaflet, a native posterior leaflet, and a native anterior leaflet of a tricuspid valve, wherein the coaptation member includes a first surface and a second surface; multiple clip mechanisms coupled to the coaptation member, wherein each of the clip mechanisms is configured to move from an open state to a closed state, wherein— in the open state, an arm member of the clip mechanism extends away from the coaptation member to allow a free end portion of the arm member to extend behind the native septal leaflet, in the closed state, the arm member presses against a ventricular side of the native septal leaflet and the first surface of the coaptation member to secure the native septal leaflet between the clip mechanism and the first surface of the coaptation member, and the second surface of the coaptation member is configured to coapt with at least one of the native anterior leaflet and the native posterior leaflet; and a stabilization member extending from the coaptation member and configured to be positioned at

least partially within a right atrium above the tricuspid valve, wherein the stabilization member is shaped to extend along and press against an atrial side of the native septal leaflet and a septal wall between the right atrium and a left atrium to (a) stabilize a position of the coaptation member relative to the tricuspid valve and (b) provide a platform for tissue ingrowth and long-term fixation, and wherein the stabilization member is biased outwardly from the coaptation member to push a portion of the native septal leaflet back from an opening of the tricuspid valve.

20. The tricuspid valve repair device of claim 19 wherein the tricuspid valve includes a flow axis, and wherein the stabilization member extends away from the coaptation member toward the septal wall at an angle relative to the flow axis.

21. The tricuspid valve repair device of claim 19 wherein the stabilization member includes a wireform extending from the coaptation member and a covering over the wireform.

22. The tricuspid valve repair device of claim 19 wherein the stabilization member is shaped to extend along and press against the atrial side of the native septal leaflet and the septal wall between the right atrium and the left atrium to substantially maintain the coaptation member in a substantially stationary position relative to the tricuspid valve during cardiac cycles.

23. A method of repairing a tricuspid valve, the method comprising: deploying a coaptation member of a tricuspid valve repair device at least partially between a first native leaflet and a second native leaflet of a tricuspid valve; grasping a first portion of a ventricular side of the first native leaflet with a first clip mechanism of the tricuspid valve repair device; and grasping a second portion of the ventricular side of the first native leaflet with a second clip mechanism of the tricuspid valve repair device to secure the first native leaflet between the first and second clip mechanisms and a first surface of the coaptation member; positioning the coaptation member such that a second surface of the coaptation member is configured to coapt with at least the second native leaflet; and contacting a stabilization member of the tricuspid valve repair device with an atrial side of the first native leaflet and a wall within a right atrium above the first native leaflet to stabilize a position of the coaptation member relative to the tricuspid valve, wherein the stabilization member is biased outwardly from the coaptation member to push a portion of the first native leaflet back from an opening of the tricuspid valve.

24. The method of claim 23 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve.

25. The method of claim 23 wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

26. The method of claim 23 wherein the one of the native leaflets is a native septal leaflet of the tricuspid valve, and wherein the wall within the right atrium is a septal wall between the right atrium and a left atrium.

27. The method of claim 23 wherein contacting the stabilization member of the tricuspid valve repair device with the atrial side of the first native leaflet and the wall within the right atrium above the first native leaflet further comprises contacting the stabilization member with the atrial side of the first native leaflet and the wall within the right atrium above the first native leaflet to substantially maintain the coaptation member in a substantially stationary position relative to the tricuspid valve during cardiac cycles.
